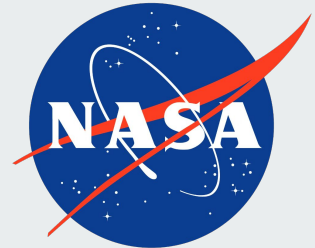




# Hands-On GIS/Remote Sensing Lab

Cloud-Streamed SST Analysis (OSTIA/MUR)



Guest Speakers: Johnathan Aguilar, Jacob Spier



# Introduction

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- Our goal is to use remote sensing data to locate places of thermal refugia
- Why? Large events of coral bleaching stresses the importance of being able to identify areas of thermal refugia
- Our study focuses on 10 years (2015-2024) of SST measurements from the Seaflower MPA
- The dataset used is the MUR SST v4 (GHR SST Level-4), produced by NASA JPL under the MEaSUREs program



QR code to Story Map

# Sign up for earthaccess #1

<https://urs.earthdata.nasa.gov/>

 EARTHDATA LOGIN

Documentation

Username 

Username is required

Password



LOG IN

 Need Help?

[Forgot Username](#) • [Forgot Password](#)

Why must I register?

The Earthdata Login provides a single mechanism for user registration and profile management for all EOSDIS system components (DAACs, Tools, Services). Your Earthdata login also helps the EOSDIS program better understand the usage of EOSDIS services to improve user experience through customization of tools and improvement of services. EOSDIS data are openly available to all and free of charge except where governed by international agreements.

REGISTER FOR A PROFILE

By clicking the Log In button above, you are acknowledging that all Earthdata Login applications running in DAACs will have access to my profile information.

# Sign up for earthaccess #2

- Be sure to remember Username and Password as these will be used in the jupyter notebook

## Register for an Earthdata Login Profile

### Profile Information

Username: •

Username is required

Password: •

Password Confirmation: •

• Required field

#### Username must:

- Be a Minimum of 4 characters
- Be a Maximum of 30 characters
- Contain at least one letter
- Use only lowercase letters, numbers, periods and underscores
- Not contain any blank spaces
- Not begin, end or contain two consecutive special characters( . \_ )

#### Password must contain:

- Minimum of 12 characters
- One Uppercase letter
- One Lowercase letter
- One Number
- One Special Character

# Sign up for earthaccess #3

- Be sure to use CSUN email

**E-mail:**

## Country Information

**Country:**

## Affiliations

**Affiliation:**

**Study Area:**

**User Type:**

**Organization:**

## Agreements

☐ Please notify me via email with important information about EOSDIS science data products (e.g. updates, new data releases, quality issues), EOSDIS applications/tools (e.g. updates, service outages), and other relevant information for users.

# Sign up for earthaccess #4

By clicking the REGISTER FOR EARTHDATA LOGIN button below, you are acknowledging that all Earthdata Login applications running in DAACs [will](#) have access to my profile information.

Protection and maintenance of user profile information is described in [NASA's Web Privacy Policy](#)

Additional information on Earthdata Login terms of service can be found [here](#)

REGISTER FOR EARTHDATA LOGIN

EDIT PROFILE

Username

Password

☒ Stay signed in (this is a private workstation)

LOG IN

[Need Help?](#)

## Why must I register?

The Earthdata Login provides a single mechanism for user registration and profile management for all EOSDIS system components (DAACs, Tools, Services). Your Earthdata login also helps the EOSDIS program better understand the usage of EOSDIS services to improve user experience through customization of tools and improvement of services. EOSDIS data are openly available to all and free of charge except where governed by international agreements.

REGISTER FOR A PROFILE

## Profile Login

**You must activate your profile to login.**

Our security guidelines requires that you activate your profile before logging in.

Click the button below to send yourself a new activation link.

SEND ACTIVATION EMAIL

# Github #1

<https://github.com/JohnathanAguilar01/geog-class-lesson>

The screenshot displays the GitHub interface for the repository 'geog-class-lesson' by user 'JohnathanAguilar01'. The repository is public and has 1 branch and 0 tags. The 'Code' button is highlighted with a red box. The commit history table shows the following commits:

| Commit Hash | Author             | Message                                     | Time       | Commits   |
|-------------|--------------------|---------------------------------------------|------------|-----------|
| ad00af      | JohnathanAguilar01 | fixed backslash error                       | 3 days ago | 6 Commits |
|             |                    | first commit                                | 3 days ago |           |
|             |                    | "finished class"                            | 3 days ago |           |
|             |                    | first commit                                | 3 days ago |           |
|             |                    | fixed backslash error                       | 3 days ago |           |
|             |                    | Added PDF                                   | 3 days ago |           |
|             |                    | Fixed some bugs encountered on google colab | 3 days ago |           |

The README section shows the title 'geog-484-class-lesson'.

# Github #2

Click “Code” and from the drop down menu click “Download ZIP”

The screenshot shows the GitHub interface for the repository 'geog-class-lesson' by user 'JohnathanAguilar01'. The repository is public and has 1 branch and 0 tags. The file list includes 'Shape\_Files', 'outputs', 'README.md', 'class\_lesson.ipynb', 'class\_lesson.pdf', and 'requirements.txt'. The 'Code' button is highlighted in green, and its dropdown menu is open, showing options for cloning (Local, Codespaces) and downloading (Download ZIP, which is highlighted with a red box). The right sidebar shows the 'About' section with no description, website, or topics provided, and the 'Releases' section with no releases published.

geog-class-lesson Public

main 1 Branch 0 Tags

Go to file Add file <> Code

JohnathanAguilar01 fixed backslash error

| File               | Commit                |
|--------------------|-----------------------|
| Shape_Files        | first commit          |
| outputs            | "finished class"      |
| README.md          | first commit          |
| class_lesson.ipynb | fixed backslash error |
| class_lesson.pdf   | Added PDF             |
| requirements.txt   | Fixed some bugs en    |

Clone

HTTPS SSH GitHub CLI

git@github.com:JohnathanAguilar01/geog-class-lesson

Use a password-protected SSH key.

Open with GitHub Desktop

Download ZIP

About

No description, website, or topics provided.

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

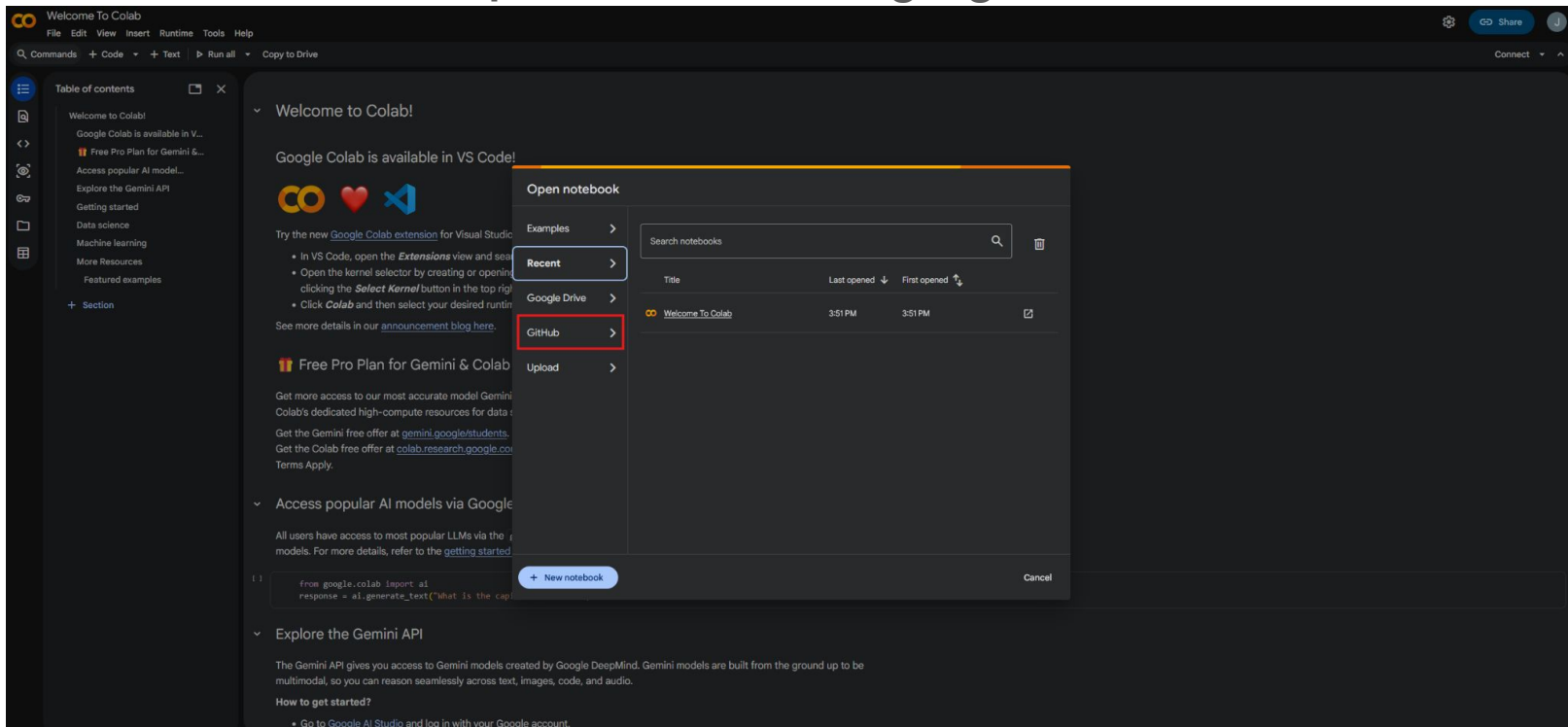
Languages

Jupyter Notebook 100.0%



# Google Collab #1

<https://colab.research.google.com/>



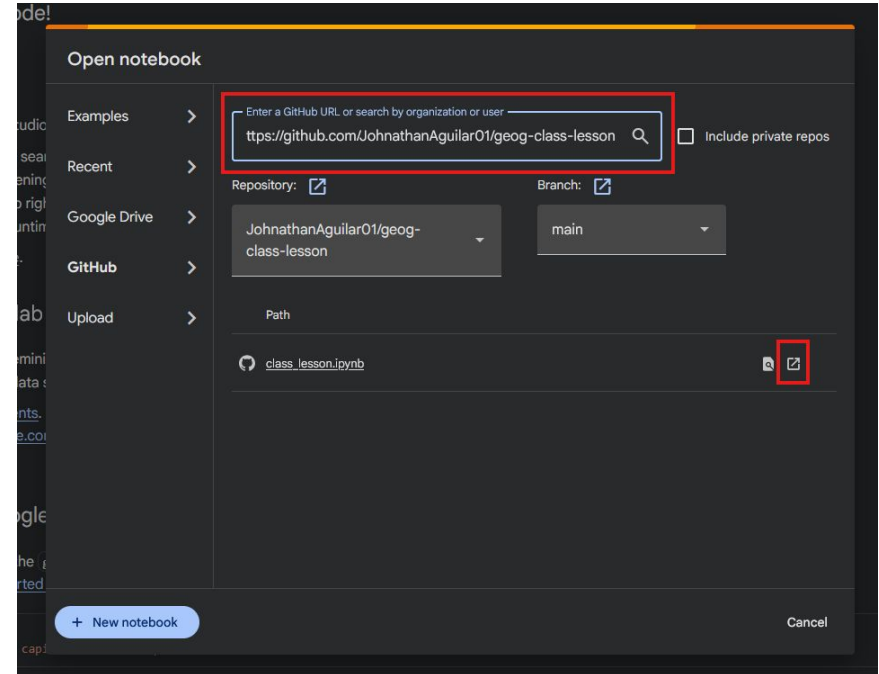
The screenshot displays the Google Colab web interface. On the left, a sidebar contains a 'Table of contents' and a list of links including 'Welcome to Colab!', 'Google Colab is available in V...', 'Free Pro Plan for Gemini &...', 'Access popular AI model...', 'Explore the Gemini API', 'Getting started', 'Data science', 'Machine learning', 'More Resources', and 'Featured examples'. The main content area shows a 'Welcome to Colab!' message with instructions on how to use Colab in VS Code, a link to the 'Free Pro Plan for Gemini & Colab', and a section for 'Access popular AI models via Google'. An 'Open notebook' dialog box is open in the center, featuring a search bar, a list of notebooks, and a 'New notebook' button. The 'Recent' tab is selected, showing a table of notebooks.

| Title            | Last opened | First opened |
|------------------|-------------|--------------|
| Welcome To Colab | 3:51 PM     | 3:51 PM      |

The 'Open notebook' dialog box also includes a sidebar with options: 'Examples', 'Recent', 'Google Drive', 'GitHub' (highlighted with a red box), and 'Upload'. At the bottom of the dialog, there is a '+ New notebook' button and a 'Cancel' button.

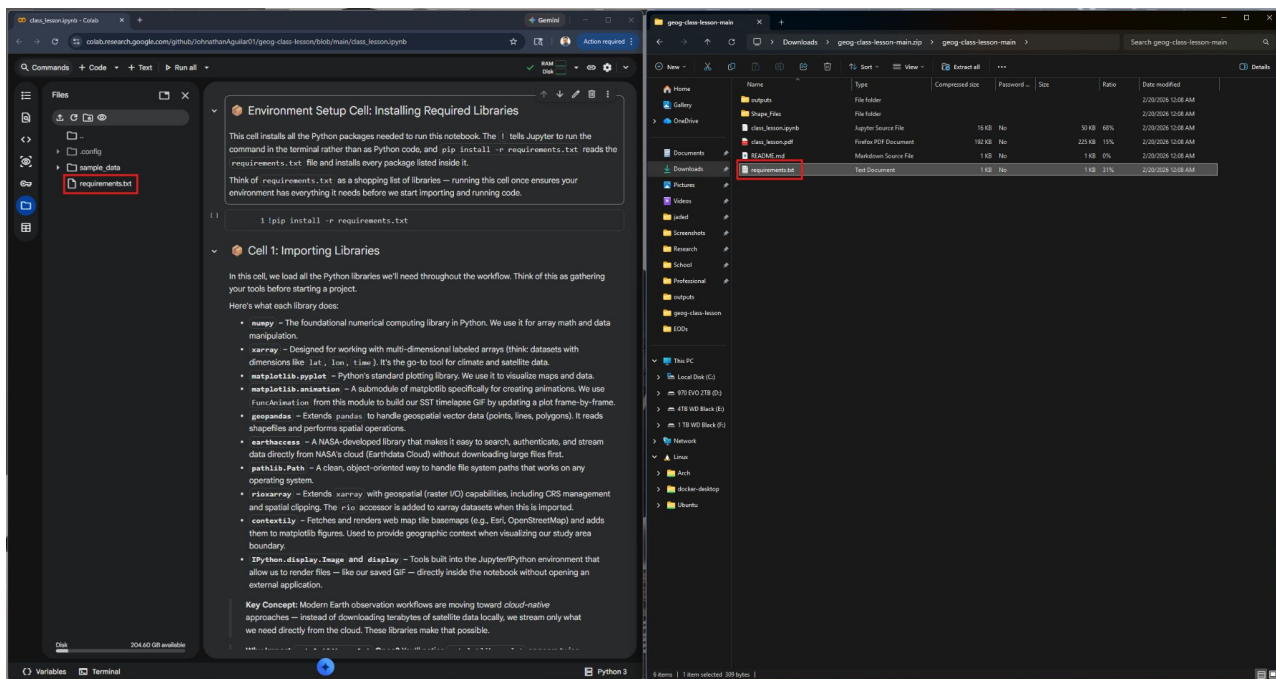
# Google Collab #2

- Enter the Github link previously used to download the ZIP file
- <https://github.com/JohnathanAguilar01/geog-class-lesson>
- Then click the button in the red box to open the Jupyter Notebook in Google Colab



# Google Collab #3

Open downloaded ZIP by double clicking, then drag and drop “requirements.txt” into Google Collab folder



# Google Collab #4

Next open the “Shape\_Files” folder and drag and drop all the contents into the Google Collab folder

The image shows two side-by-side windows. The left window is a Google Colab notebook titled "class\_lesson.ipynb". The file browser on the left shows a folder named "Shape\_Files" containing several files: "Caribbean\_Study\_Area.csg", "Caribbean\_Study\_Area.dxf", "Caribbean\_Study\_Area.prj", "Caribbean\_Study\_Area.shp", "Caribbean\_Study\_Area.shx", "Caribbean\_Study\_Area.spr", "Caribbean\_Study\_Area.tbx", and "requirements.txt". The main content area shows "Environment Setup Cell: Installing Required Libraries" with a code cell containing `!pip install -r requirements.txt`. Below this is "Cell 1: Importing Libraries" with a list of libraries and their descriptions.

The right window is a Windows File Explorer showing the "Shape\_Files" folder. It contains a table of files with columns: Name, Type, Compressed size, Password, Size, Ratio, and Date modified. The files listed are:

| Name                     | Type                         | Compressed size | Password | Size  | Ratio | Date modified      |
|--------------------------|------------------------------|-----------------|----------|-------|-------|--------------------|
| Caribbean_Study_Area.csg | CAD File                     | 1 KB            | No       | 1 KB  | 0%    | 2/28/2024 10:08 AM |
| Caribbean_Study_Area.dxf | DWG File                     | 1 KB            | No       | 1 KB  | 44%   | 2/28/2024 10:08 AM |
| Caribbean_Study_Area.prj | PRJ File                     | 1 KB            | No       | 1 KB  | 17%   | 2/28/2024 10:08 AM |
| Caribbean_Study_Area.shp | SHX File                     | 1 KB            | No       | 1 KB  | 56%   | 2/28/2024 10:08 AM |
| Caribbean_Study_Area.shx | SHX File                     | 1 KB            | No       | 1 KB  | 15%   | 2/28/2024 10:08 AM |
| Caribbean_Study_Area.spr | SHP File                     | 10 KB           | No       | 25 KB | 34%   | 2/28/2024 10:08 AM |
| Caribbean_Study_Area.tbx | Microsoft Edge HTML Document | 3 KB            | No       | 7 KB  | 63%   | 2/28/2024 10:08 AM |
| Caribbean_Study_Area.shp | SHX File                     | 1 KB            | No       | 1 KB  | 15%   | 2/28/2024 10:08 AM |